

Many polynomial-size dice would settle the exponential-total question

Asymptotic reductions from hereditary fairness

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Let A_r be the minimum total number of faces of a permutation-fair family of r finite equiprobable dice, with arbitrary individual sizes. All logarithms below are natural.

Proposition 1 (Arithmetic upper bound on the exceptional set). *If a permutation-fair family contains r dice with at most M faces each, then, as $r \rightarrow \infty$,*

$$\log M \geq (1/2 + o(1))r.$$

Consequently an N -die family contains at most $(2C + o(1)) \log N$ dice with at most N^C faces, for fixed $C > 0$.

Proof. Restrict to the r specified dice, with counts $K_1, \dots, K_r$. For every prime $p \leq r$, at most $p - 1$ of these counts can fail to be divisible by p : otherwise a p -element restriction would have a face-count product not divisible by $p!$. Thus

$$r \log M \geq \sum_{i=1}^r \log K_i \geq \sum_{p \leq r} (r - p + 1) \log p.$$

Writing $\vartheta(x) = \sum_{p \leq x} \log p$, the last sum is $\vartheta(r) + \int_2^r \vartheta(t) dt = (1/2 + o(1))r^2$ by the prime number theorem. This proves the claims; a bounded r is harmless in the second asymptotic assertion. $\square$

Proposition 2 (Reduction to the first open problem). *Suppose there are absolute constants $c, C > 0$ such that, for every sufficiently large integer N , some permutation-fair N -die family contains at least $c \log N$ dice with at most N^C faces each. Then*

$$A_r \leq \exp(O(r)) \quad (r \rightarrow \infty).$$

More precisely $\log A_r \leq (C/c + o(1))r$.

Proof. For each large r , take $N = \lceil \exp((r+1)/c) \rceil$ and retain any r of the promised small dice. Hereditary fairness gives a fair r -die family of total size at most rN^C . Its logarithm is at most $(C/c)r + O(\log r)$. $\square$

Interpretation and limits. The first proposition is a sharpening of the manuscript's arithmetic maximum-die argument, not a new construction. The second is a one-way reduction. A proof of logarithmically many uniformly polynomial-size exceptional dice would already close the original exponential versus $\exp(O(r \log r))$ upper-bound gap. Conversely, an exponential-total construction for r dice does not automatically embed into a much larger fair family while keeping those dice small; no equivalence is claimed.

The proved fixed- k construction, including its uniform quantitative bound

$$S \leq (N + 2^k + 2)^{\exp(Ck^2)}$$

for k specified equal-count dice, does not meet the uniform hypotheses of Proposition 2. Its growing- k consequences give subexponential sizes, not a bound N^C with an absolute exponent. See `size_bounds/uniform_fixed_number_complexity.tex`.